

# Removing Batch Effects from Genomics Data

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# ComBat Intuition

Consider the following model:

$$Y_{ijg} = \alpha_g + X\beta_g + \gamma_{ig} + \delta_{ig}\epsilon_{ijg}$$

where:

- ▶  $\alpha_g$  is the overall gene expression
- ▶  $X$  is a design matrix
- ▶  $\beta_g$  contains the regression coefficients
- ▶ The error terms  $\epsilon_{ijg} \sim N(0, \sigma_g^2)$
- ▶  $\gamma_{ig}$  and  $\delta_{ig}$  are additive and multiplicative batch effects

(Johnson et al., Biostatistics, 2007)}}}

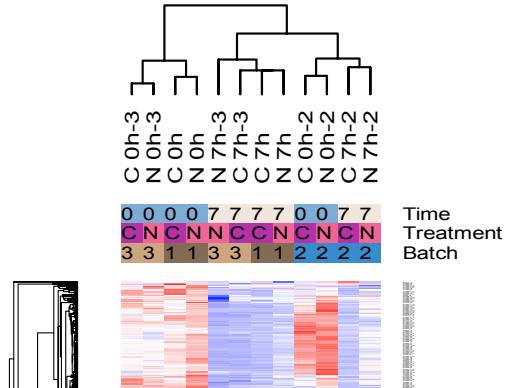
# Batch effects

**Batch Effect:** Non-biological variation due to differences in batches of data that confound the relationships between covariates of interest.

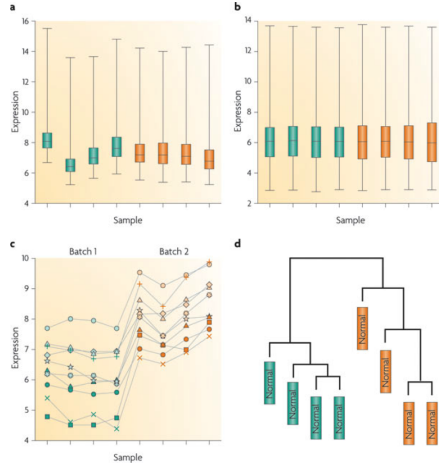
Caused by differences:

- ▶ Gene expression profiling platform
- ▶ Lab protocol or experimenter
- ▶ Time of day or hybridization
- ▶ Atmospheric ozone level (Rhodes et al. 2004)

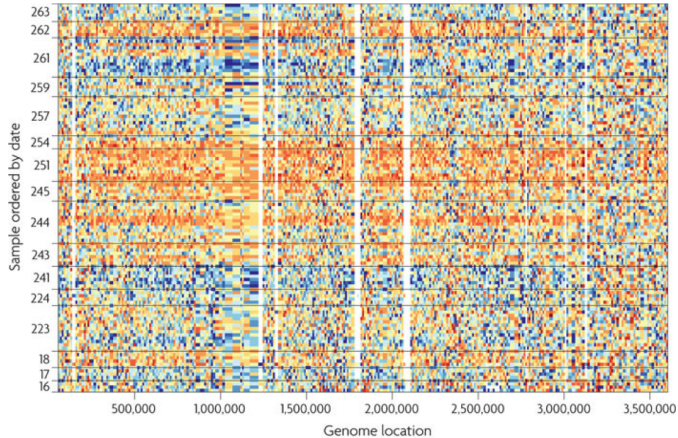
# Batch effect examples: Nitric Oxide



# Batch effect examples: Bladder cancer



# Batch effect examples: 1000 genomes



# Batch effect examples: Proteomics

Proteomic data with batch effects:

- ▶ Proteomic markers to predict endometriosis (39 total)
- ▶ Single peptide predictors of disease (AUC): 0.82, 0.76, 0.74, 0.74, 0.70 (+12 more  $\geq 0.6$ )
- ▶ Single peptide predictors of batch (AUC): 0.99, 0.94, 0.91, 0.86, 0.86, 0.84, 0.84, 0.84, 0.83, 0.82 (+7 more  $\geq 0.6$ )
- ▶ Predict batch better than disease!

# Normalization?

- ▶ **Question:** Shouldn't normalization take care of this?
  - ▶ Answer: NO!
- ▶ Batch effects often impacts genes or sets of genes
- ▶ Batch effects often remain after normalization



## Adjusting for batch effects

# Early methods for batch effects

- ▶ Singular value decomposition (Alter et al., 2000)
- ▶ Distance weighed discrimination (Benito et al., 2004)
- ▶ ComBat (Johnson et al., 2007)
- ▶ Surrogate variable analysis (Leek and Storey, 2007)
- ▶ Cross Platform Normalization (Shabaline et al., 2008)
- ▶ Barcoding (Zilliox and Irizarry 2007; Piccolo et al. 2013)
- ▶ Single sample normalization (Hubbell et al., 2002; McCall et al., 2010; Piccolo et al. 2012)
- ▶ Removing unwanted variation (Risso et al., 2014; Jacob et al., 2016)

# ComBat Intuition

Consider the following model:

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# ComBat Intuition

Adjust for batch effects:

$$Y_{ijg}^* = \frac{Y_{ig} - \hat{\alpha}_g - X\hat{\beta}_g - \hat{\gamma}_{ig}}{\hat{\delta}_{ig}} + \hat{\alpha}_g + X\hat{\beta}_g$$

**Problem:** How to robustly estimate the parameters?

**Answer:** Empirical Bayes!

# Step 1: Standardize the data

Genes are on different scales, so first standardize the data:

$$Z_{ijg} = \frac{Y_{ijg} - \hat{\alpha}_g - X\hat{\beta}_g}{\hat{\sigma}_g}$$

where  $\hat{\alpha}_g$ ,  $\hat{\beta}_g$ , and  $\hat{\sigma}_g^2$  are estimated using gene-wise MLEs.

## Step 2: EB batch effect estimates

**Additive batch adjustment ( $\gamma_{ig}$ ):** Using Bayes theorem and a *Gaussian*( $\gamma_i, \tau_i^2$ ) conjugate prior:

$$E[\gamma_{ig} | \mathbf{Z}_{ig}, \delta_{ig}^2] = \frac{\tau_i^2 \sum_j Z_{ijg} + \delta_{ig}^2 \gamma_i}{n_i \tau_i^2 + \delta_{ig}^2}.$$

which can therefore be estimated as

$$\gamma_{ig}^* = \hat{E}[\gamma_{ig} | \mathbf{Z}_{ig}, \delta_{ig}^2] = \frac{n_i \bar{\tau}_i^2 \hat{\gamma}_{ig} + \delta_{ig}^{2*} \bar{\gamma}_i}{n_i \bar{\tau}_i^2 + \delta_{ig}^{2*}}.$$

## Step 2: EB batch effect estimates

**Multiplicative batch adjustment ( $\delta_{ig}$ ):** Using Bayes theorem and an *Inverse Gamma*( $\lambda_i, \theta_i$ ) conjugate prior:

$$E[\delta_{ig}^2 | \mathbf{Z}_{ig}, \gamma_{ig}] = \frac{\theta_i + \frac{1}{2} \sum_j (Z_{ijg} - \gamma_{ig})^2}{\frac{n_i}{2} + \lambda_i - 1}.$$

which can therefore be estimated as

$$\delta_{ig}^{2*} = \hat{E}[\delta_{ig}^2 | \mathbf{Z}_{ig}, \gamma_{ig}] = \frac{\bar{\theta}_i + \frac{1}{2} \sum_j (Z_{ijg} - \gamma_{ig}^*)^2}{\frac{n_i}{2} + \bar{\lambda}_i - 1}.$$

## Step 2: EB batch effect estimates

Estimate hyperpriors using the Method of Moments across all genes:

For the additive batch adjustment:

$$\bar{\gamma}_i = \frac{1}{G} \sum_g \hat{\gamma}_{ig}, \text{ and } \bar{\tau}_i^2 = \frac{1}{G-1} \sum_g (\gamma_{ig} - \bar{\gamma}_i)^2.$$

For the multiplicative batch adjustment:

$$\bar{\lambda}_i = \frac{\bar{V}_i + 2\bar{S}_i^2}{\bar{S}_i^2} \text{ and } \bar{\theta}_i = \frac{\bar{V}_i^3 + \bar{V}_i\bar{S}_i^2}{\bar{S}_i^2}.$$



## Step 2: EB batch effect estimates

Alternative non-parametric prior, assume:

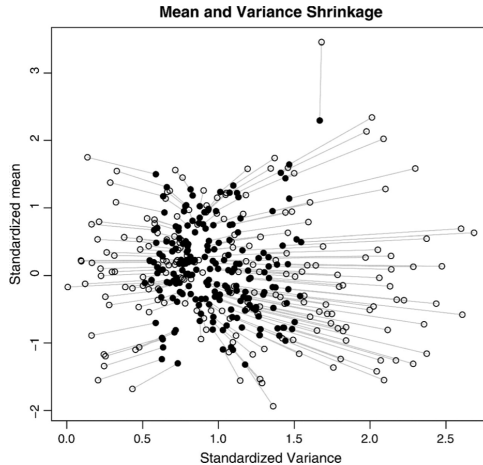
$$\mathbf{Z}_{ijg} \sim N(\gamma_{ig}, \delta_{ig}^2), \text{ and let } w_{igk} = L(\mathbf{Z}_{ig} | \hat{\gamma}_{ik}, \hat{\delta}_{ik}^2).$$

The nonparametric EB batch adjustments  $\gamma_{ig}^*, \delta_{ig}^{2*}$  are given by

$$\gamma_{ig}^* = \frac{\sum_k w_{igk} \hat{\gamma}_{ik}}{\sum_k w_{igk}} \text{ and } \delta_{ig}^{2*} = \frac{\sum_k w_{igk} \hat{\delta}_{ik}^2}{\sum_k w_{igk}},$$

i.e. Monte Carlo integration over an unspecified empirical prior.

# Empirical Bayes Shrinkage

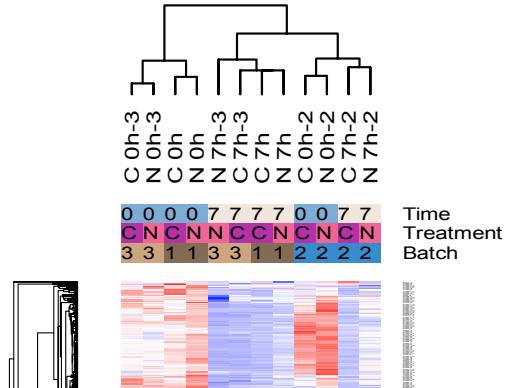


## Step 3: Adjust the data for batch effects

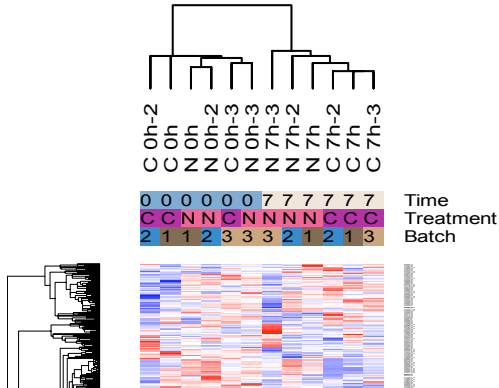
The EB batch adjusted data  $Y_{ijg}^*$  can be calculated using EB estimated batch effects:

$$Y_{ijg}^* = \frac{\hat{\sigma}_g}{\delta_{ig}^*} (Z_{ijg} - \gamma_{ig}^*) + \hat{\alpha}_g + X\hat{\beta}_g.$$

# Batch effect examples: Nitric Oxide



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# Batch effect examples: Proteomics

Proteomics data with batch effects:

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- ▶ Predict batch better than disease!

# Batch effect examples: Proteomics

After ComBat batch adjustment:

- ▶ Single peptide predictors of disease (AUC): 0.80, 0.79, 0.75, 0.70, 0.70 (+7 more  $\geq 0.6$ )
- ▶ Single peptide predictors of batch (AUC): 0.64, 0.60

## Confounded Designs

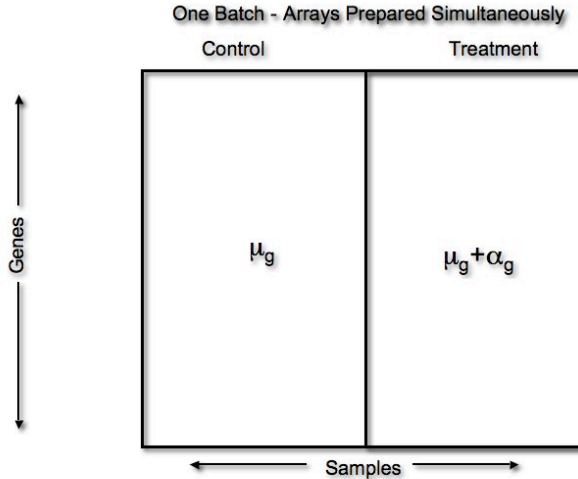


# Confounded Designs

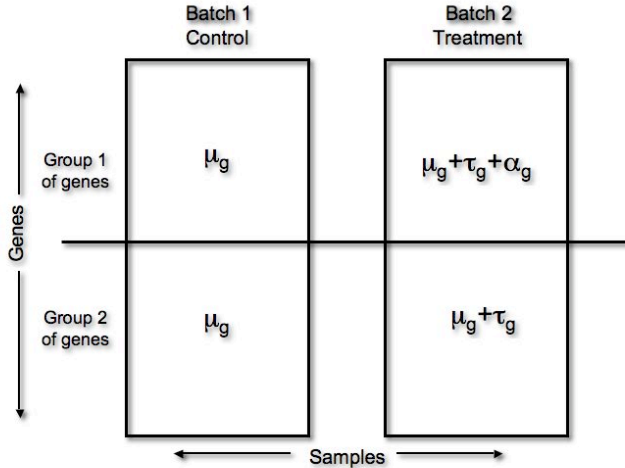
What do you do in the following cases?

- ▶ Treatments in one batch, controls in another
- ▶ Treatment 1 in Batch 1, Treatment 2 in Batch 2
- ▶ Same experiment with different tissues or cell lines
- ▶ Some samples are in their own batch?
- ▶ Can I do with anything with these cases?

# Confounded Designs



# Confounded Designs—Solution!



# Confounded Designs—Solution!

Assumptions for parametric ComBat: 1. Gaussian data:

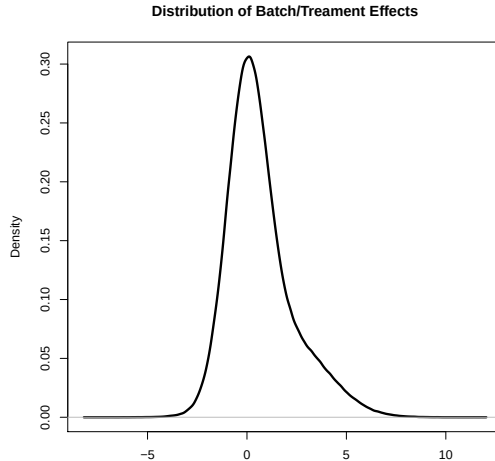
$Z_{ijg} \sim N(\gamma_{ig}, \delta_{ig}^2)$  2. Systematic batch effects:  $\gamma_{ig} \sim N(\gamma_i, \tau_i^2)$ ,  
 $\delta_{ig}^2 \sim IG(\lambda_i, \theta_i)$

Assumptions for confounded ComBat: 1.  $Z_{ijg} \sim N(\gamma_{ig}, \delta_{ig}^2)$  2.

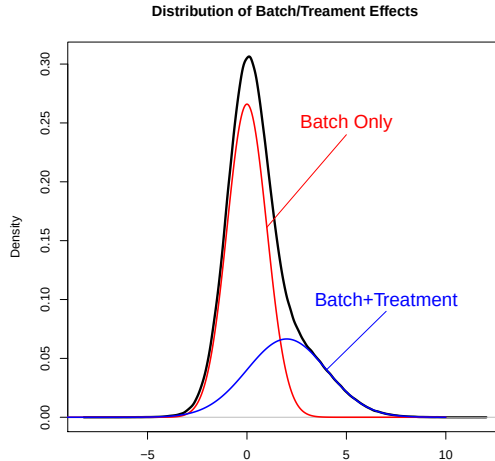
Some (but not all) have treatment effects:  $\phi_g \sim \text{Bern}(\pi)$  3.

Systematic:  $\gamma_{ig} \sim N(\gamma_i + \phi_g \eta_i, \tau_i^2 + \phi_g \rho_i^2)$ ,  $\delta_{ig}^2 \sim IG(\lambda_i, \theta_i)$

# Confounded Designs—Solution!



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# Confounded Designs—Solution!

Confounded ComBat steps:

1. Standardize the data
2. Estimation of the hyper priors via the EM algorithm
3. EB batch/treatment estimates using maximum posterior estimates
4. Adjust the data and restore original scaling

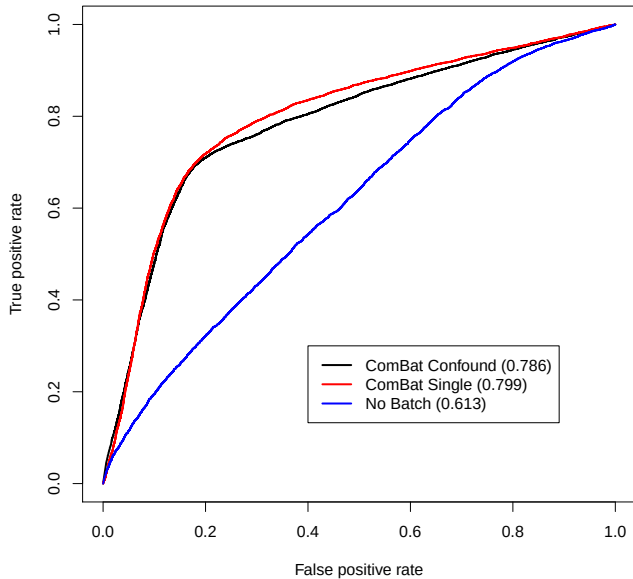
**Note:** This works on single samples as well!

# RAS and PI3K Pathway Activation

- ▶ Human primary mammary epithelial cells
- ▶ Transfect with an adenovirus expressing RAS or PI3K
- ▶ Observe change compared to control
- ▶ RAS and PI3K profiled at different times on different platforms
- ▶ Control cells are the same!



# PI3K vs. RAS



# RAS and PI3K Pathway Activation

Results with p-value cutoff of 0.001:

|                 | PI3K vs. RAS |       |       |
|-----------------|--------------|-------|-------|
|                 | Sens         | Spec  | AUC   |
| ComBat Confound | 0.747        | 0.730 | 0.786 |
| ComBat Single   | 0.764        | 0.740 | 0.799 |
| No Batch        | 0.787        | 0.361 | 0.613 |

**Confounding:** decrease in sensitivity (25-29%) and specificity (22-26%)

# Session Info

```
sessionInfo()
```

```
## R version 4.5.2 (2025-10-31)
## Platform: aarch64-apple-darwin20
## Running under: macOS Tahoe 26.5.1
##
## Matrix products: default
## BLAS:   /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/libBLAS.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib; LAPACK version 3.12.1
##
## locale:
## [1] en_US/en_US/en_US/C/en_US/en_US
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods   base
##
## loaded via a namespace (and not attached):
## [1] compiler_4.5.2 fastmap_1.2.0 cli_3.6.6      tools_4.5.2
## [5] htmltools_0.5.9 otel_0.2.0    yaml_2.3.12   rmarkdown_2.31
## [9] knitr_1.51      xfun_0.58     digest_0.6.39 rlang_1.2.0
## [13] evaluate_1.0.5
```